



Cardiovascular Disease

Cardiovascular diseases (CVDs) affect your heart and blood vessels. Almost half of all adults in the U.S. have at least one form of heart disease. You may make lifestyle changes to manage cardiovascular disease or your healthcare provider may prescribe medications. The sooner you detect cardiovascular disease, the easier it is to treat.

Overview

What is cardiovascular disease?

Cardiovascular disease is a group of diseases affecting your heart and blood vessels. These diseases can affect one or many parts of your heart and/or blood vessels. A person may be symptomatic (physically experiencing the disease) or asymptomatic (not feeling anything at all).

Cardiovascular disease includes heart or blood vessel issues, including:

- Narrowing of the blood vessels in your heart, other organs or throughout your body.
- Heart and blood vessel problems present at birth.
- Heart valves that aren't working right.
- Irregular heart rhythms.

How common is cardiovascular disease?

Cardiovascular disease is the leading cause of death worldwide and in the U.S.

Almost half of adults in the U.S. have some form of cardiovascular disease. It affects people of all ages, **sexes**, ethnicities and socioeconomic levels. One in three women dies from cardiovascular disease.

Signs and Symptoms

What are the symptoms of cardiovascular disease?

Cardiovascular disease symptoms can vary depending on the cause. Older adults and women may have more subtle symptoms. However, they can still have serious cardiovascular disease.

Symptoms of heart issues

- Chest pain (angina).
- Chest pressure, heaviness or discomfort, sometimes described as a "belt around the chest" or a "weight on the chest."
- Shortness of breath (dyspnea).
- Dizziness or fainting.

- Fatigue or exhaustion.

Symptoms of blockages in blood vessels throughout your body

- Pain or cramps in your legs when you walk.
- Leg sores that aren't healing.
- Cool or red skin on your legs.
- Swelling in your legs.
- Numbness in your face or a limb. This may be on only one side of your body.
- Difficulty with talking, seeing or walking.

What conditions are cardiovascular diseases?

There are many different types of cardiovascular diseases, including but not limited to:

- **Arrhythmia:** Problem with your heart's electrical conduction system, which can lead to abnormal heart rhythms or heart rates.
- **Valve disease:** Tightening or leaking in your heart valves (structures that allow blood to flow from one chamber to another chamber or blood vessel).
- **Coronary artery disease:** Problem with your heart's blood vessels, such as blockages.
- **Heart failure:** Problem with heart pumping/relaxing functions, leading to fluid buildup and shortness of breath.
- **Peripheral artery disease:** Issue with the blood vessels of your arms, legs or abdominal organs, such as narrowing or blockages.
- **Aortic disease:** Problem with the large blood vessel that directs blood from your heart to your brain and the rest of your body, such as dilatation or aneurysm.
- **Congenital heart disease:** Heart issue that you're born with, which can affect different parts of your heart.
- **Pericardial disease:** Problem with the lining of your heart, including pericarditis and pericardial effusion.
- **Cerebrovascular disease:** Issue with the blood vessels that deliver blood to your brain, such as narrowing or blockages.
- **Deep vein thrombosis (DVT):** Blockage in your veins, vessels that bring blood back from your brain/body to your heart.

Causes

What causes cardiovascular disease?

The causes of cardiovascular disease can vary depending on the specific type. For example, atherosclerosis (plaque buildup in your arteries) causes coronary artery disease and peripheral artery disease. Coronary artery disease, scarring of your heart muscle, genetic problems or medications can cause arrhythmias. Aging, infections and rheumatic disease can cause valve diseases.

What are cardiovascular disease risk factors?

You may be more likely to develop cardiovascular disease if you have risk factors such as:

- High blood pressure (hypertension).
- High cholesterol (hyperlipidemia).
- Tobacco use (including vaping).
- Type 2 diabetes.
- Family history of heart disease.
- Lack of physical activity.
- Having excess weight or obesity.
- Diet high in sodium, sugar and fat.
- Overuse of alcohol.
- Misuse of prescription or recreational drugs.
- Preeclampsia or toxemia.
- Gestational diabetes.
- Chronic inflammatory or autoimmune conditions.
- Chronic kidney disease.

Diagnosis and Tests

How is cardiovascular disease diagnosed?

Your healthcare provider will perform a physical exam and ask questions about your symptoms, personal health and family health history. They may also order tests to help diagnose cardiovascular disease.

What tests might I have for cardiovascular disease?

Some common tests to diagnose cardiovascular disease include:

- **Blood work** measures substances that indicate cardiovascular health, such as cholesterol, blood sugar levels and specific proteins. A provider can use a blood test to check for blood clotting issues as well.
- **Ankle brachial index (ABI)** compares the blood pressure in your ankles and arms to diagnose peripheral artery disease.
- **Electrocardiogram (EKG)** records your heart's electrical activity.
- **Ambulatory monitoring** uses wearable devices that track your heart rhythm and rates.
- **Echocardiogram** uses sound waves to create an image of your heartbeat and blood flow.
- **Ultrasound** uses sound waves to check blood flow in your legs or neck.
- **Cardiac computerized tomography (CT)** uses X-rays and computer processing to create 3D images of your heart and blood vessels.
- **Cardiac magnetic resonance imaging (MRI)** uses magnets and radio waves to create highly detailed images of your heart.
- **MR angiogram or CT angiogram** uses an MRI or CT, respectively, to see blood vessels in your legs, head and neck.
- **Stress tests** analyze how physical activity affects your heart in a controlled setting, using exercise or medications, to determine how your heart responds. This type of test can involve EKGs and/or imaging tests.

- **Cardiac catheterization** uses a catheter (thin, hollow tube) to measure pressure and blood flow in your heart.

Management and Treatment

How is cardiovascular disease treated?

Treatment plans can vary depending on your symptoms and the type of cardiovascular disease you have. Cardiovascular disease treatment may include:

- **Lifestyle changes:** Examples include changing your diet, increasing your aerobic activity and quitting smoking or tobacco products (including vaping).
- **Medications:** Your healthcare provider may prescribe medications to help manage cardiovascular disease. Medication type will depend on what kind of cardiovascular disease you have.
- **Procedures or surgeries:** If medications aren't enough, your healthcare provider may use certain procedures or surgeries to treat your cardiovascular disease. Examples include **stents** in your heart or leg arteries, **minimally invasive heart surgery**, **open-heart surgery**, **ablations** or **cardioversion**.
- **Cardiac rehabilitation:** You may need a monitored exercise program to help your heart get stronger.
- **Active surveillance:** You may need careful monitoring over time without medications or procedures/surgeries.

Outlook / Prognosis

What is the outlook for people with cardiovascular disease?

Many people enjoy a high quality of life and can manage their cardiovascular disease with the help of their healthcare team. Your chances for a positive outcome are higher if you engage in your healthcare and follow your provider's treatment plan. It's important to take medications exactly as prescribed.

Does cardiovascular disease increase my risk of other conditions?

Untreated cardiovascular disease can lead to serious complications.

If you have cardiovascular disease, you may have a higher risk of:

- Heart attack.
- Stroke.
- Acute limb ischemia (sudden blockage in your leg arteries).
- Aortic dissection.
- Sudden cardiac death.

Prevention

How can I prevent cardiovascular disease?

You can't prevent some types of cardiovascular disease, such as congenital heart disease. But lifestyle changes can reduce your risk of many types of cardiovascular disease.

You can reduce your cardiovascular risks by:

- Avoiding all tobacco products.
- Managing other health conditions, such as Type 2 diabetes, high cholesterol or high blood pressure.
- Achieving and maintaining a healthy weight.
- Eating a diet low in saturated fat and sodium.
- Exercising at least 30 to 60 minutes per day on most days.
- Reducing and managing stress.

Living With

When should I see my healthcare provider?

Cardiovascular disease is often easier to treat when healthcare providers catch it early. That's why it's important to see a primary care provider every year. They can detect cardiovascular issues before symptoms start. If you have any signs of cardiovascular disease, you should see your provider immediately.

Call 911 or seek emergency medical attention if you experience sudden:

- Chest pain, pressure, heaviness or discomfort, especially with exertion.
- Fainting (syncope).
- Severe shortness of breath, especially if it's new or progressive.
- Pain or numbness in your arms/legs.
- Ripping or tearing back pain.

A note from Cleveland Clinic

Cardiovascular diseases are conditions that affect your heart and blood vessels. Without appropriate treatment, heart disease can lead to heart attacks or strokes. You can make lifestyle changes or take medications to manage cardiovascular disease. Earlier diagnosis can help with effective treatment. Many people live a full and active life with a cardiovascular disease.

Experts You Can Trust

✓ **Medically Reviewed.** Last updated on 09/01/2022.

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Cleveland Clinic's health articles are based on evidence-backed information and review by medical professionals to ensure accuracy, reliability and up-to-date clinical standards.

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